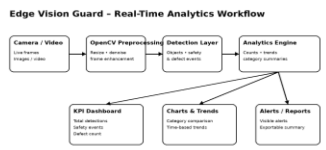
SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai - 602105

Student Name:	Rakina S	Reg. No.:	192324289
Course Code:	DSA0202	Slot:	D
Course Name:	Computer Vision with OpenCV		
Course Faculty:	Dr M Mayuranathan		

Project Title: EDGE VISION GUARD: REAL-TIME SAFETY & DEFECT ANALYTICS SYSTEM

<table>
<tr><th>Module Photo</th><th>Individual student contribution module work in the project</th><th>Presentation Image</th></tr>
<tr><td>Module 3 – Real-Time Analytics & Visualization</td><td>Implemented the real-time analytics and visualization workflow: collecting detection outputs, calculating safety and defect counts, comparing categories, analyzing trends, and presenting results through KPIs, charts, alerts, and a dashboard.</td><td>


Module 3: Real-time analytics and visualization architecture

</td></tr>
</table>

Module 3 – Real-Time Analytics & Visualization

The Edge Vision Guard system processes camera, video and image data for real-time safety monitoring and product-defect analytics. Module 3 focuses on converting detection results into meaningful analytics, trends and dashboard visualizations.

Module 3 Objectives:

✓ Display real-time detection results.

✓ Calculate safety-event and defect counts.

✓ Compare defect categories and safety events.

✓ Generate charts, KPIs and visual summaries.

Implementation:

The module follows the pipeline: detection output → data analysis → count calculation → visualization → dashboard display. Pandas and Matplotlib are used for data processing and visualization.

Significance & Scope:

This module provides the analytics and visualization layer of Edge Vision Guard. It helps users monitor results, identify trends and understand safety and defect information quickly.

Student Signature **Guide Signature**